

Chapter 23

Floating Tones

Nicholas Rolle 

Princeton University

Larry M. Hyman 

University of California, Berkeley

Florian Lionnet 

Princeton University

Floating tones are tones that are not associated with any tone-bearing unit (e.g., segmental material), and they have been used widely in the analysis of a variety of tone systems since the second half of the 20th century. Floating tones are inherently abstract and do not constitute an empirical object (one cannot hear a floating tone). Rather they are an analytical device, one that we show is indispensable for tonal analysis. This chapter gives an overview of floating tones, their nature, and the variety of tonal phenomena they have been used to account for (downstep, blocking effects, etc.). While most of our examples come from African languages (primarily from the Niger-Congo stock, or those within its orbit), our discussion should be relevant to tone languages in general.

1 Introduction

Tone languages can be defined as those where “an indication of pitch enters into the lexical realization of at least some morphemes” (Hyman 2001: 1368; see also Welmers 1959, 1973), which constitute roughly half the world’s languages and 80% of the languages of Africa (Maddieson 2022, 2023).¹ Tone languages typically

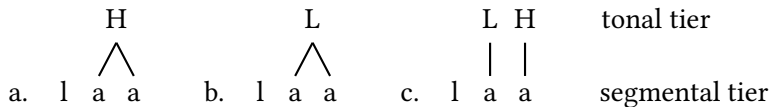
¹Maddieson bases his estimate of 46.7% on 6,520 languages listed in the *Ethnologue* (Eberhard et al. 2023) and other sources. Our 80% African calculation is based on Maddieson’s Table 2 which

possess minimal pairs such as in (1) from the Mande language Vai [vai] (Welmers 1976: 31). Here, the same segmental sequence *laa* expresses three distinct nouns depending on whether it is high (H), low (L), or rising (rendered LH). Note that *aa* stands for a long vowel, i.e., the vowel /a/ linked to two morae, that is, two Tone Bearing Units (TBUs).

- (1) Vai – Tonal minimal triplet in nouns
- | | | | | |
|----|-----------|------------|-------|---------------|
| a. | High H | <i>láá</i> | [lá:] | ‘a plankfish’ |
| b. | Low L | <i>làà</i> | [là:] | ‘a paddle’ |
| c. | Rising LH | <i>làá</i> | [lǎ:] | ‘a clay pot’ |

Perhaps the most insightful development in the analysis of tone is separating segments and tones onto separate ‘tiers’, as developed in Goldsmith’s (1976) Autosegmental Phonology. In this framework, the minimal triplet in (1) would be rendered as in (2) where phonological elements on the tonal tier are connected via ‘association lines’ to TBUs, here conflated with segmental representations on the segmental tier.

- (2) Vai – Autosegmental diagrams



Separating the tonal and segmental tiers allows one to easily model one-to-one associations (as in 2c), and also many-to-one associations where an element on one tier associates to multiple elements on another tier (2a–2b). Also, because the tiers are independent, it is possible for elements to be present on one tier without any elements occurring on the other. Thus, representations of underlying morphemes as well as surface forms may contain ‘toneless’ segments which are not associated to a tone, or tones which are not associated to a segment. The latter constitutes what is known as ‘floating tone’.

Floating tones can be demonstrated with the definite versions of these Vai nouns, shown in (3). Descriptively, this involves the addition of a rising tone at the right edge of the noun.

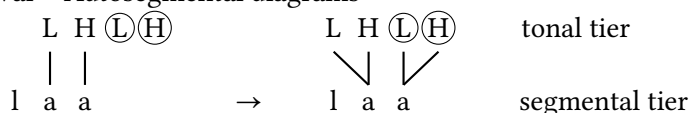
indicates the number of tonal languages in Niger-Congo (1,458 out of 1,551), Afroasiatic (252 out of 384), and Nilo-Saharan (180 out of 207), to which we added 27 Khoisan languages occurring in the *Ethnologue*.

(3) Vai – Definite forms of nouns

- a. High H láá ‘a plankfish’ láă ‘the plankfish’
 b. Low L làà ‘a paddle’ làă ‘the paddle’
 c. Rising LH làá ‘a clay pot’ láă ‘the clay pot’

We might think of the definite marker as a floating rising tone morpheme, which must associate to the segmental tier since it is not pre-associated to anything. A mapping from the underlying form to the surface form for [láă] ‘the clay pot’ is provided in (4). We follow the common convention of circling floating tones (e.g., Pulleyblank 1986, Yip 2002), thus the floating rise is represented as a $\textcircled{\text{L}}\textcircled{\text{H}}$ sequence.²

(4) Vai – Autosegmental diagrams



In what follows, we shall discuss cases akin to Vai where floating tones are part of the underlying representation of morphemes which usually associate in the course of a phonological derivation. However, there are also cases where they have been posited to remain floating either as a trigger (e.g., of downstep) or a blocker (e.g., of tone spreading), among other effects. We shall collectively refer to these as ‘floating tone effects’, and demonstrate how floating tones have proven useful in individual instances, both for underlying and derived floating tones.

The outline of our chapter is as follows. Section 2 discusses floating tones as part of morphemes or as independent morphemes themselves, which we refer to as their ‘morphological status’. Section 3 demonstrates the use of floating tones as an analytic device to account for certain tonal behavior, which we collectively call ‘floating tone effects’. This includes blocking effects, downstep and upstep/raising, and contrastive final level L tones. Section 4 discusses certain analytical issues which come up when one adopts floating tones, specifically related to the inventory of floating tones, and issues of locality, prosodic category, directionality, and markedness. Section 5 provides several analytical alternatives to floating tones, and briefly assesses their respective merits. Section 6 provides our conclusion.

²Although the $\textcircled{\text{L}}\textcircled{\text{H}}$ input is no longer floating in the output, we leave the circles around the tones for identificational purposes.

2 Morphological status

Given autosegmental architecture, the only difference between floating tones and non-floating tones is that the former are unassociated to a host. As such, we expect floating tones to have (roughly) the same morphological status as their associated counterparts. In this section, we show how floating tones may be the sole expression of the morpheme, referred to as ‘tonal morphemes’ (Section 2.1), or accompany the segmental components of a morpheme, where they act as part of the expression of the morpheme’s meaning (Section 2.2).

2.1 Tonal morphemes

Cases where tone alone expresses linguistic meaning are referred to as ‘tonal morphemes’. We have seen this already in Vai (3–4), where the marker of definiteness was said to be a floating tone sequence $\textcircled{\text{L}}\textcircled{\text{H}}$ which docks to the right edge of a noun. Such tonal morphemes are capable of a full range of functions on par with their segmental counterparts, marking both derivational meanings (e.g., changes to part-of-speech, valency, diminutivization, *inter alia*) and inflectional meanings (e.g., case, number, gender/class, agreement/concord, finiteness, tense/aspect/mood, polarity/negation, realis, focus/topic, *inter alia*). In fact, the only type of tonal morpheme we have not seen is that of a lexical root. One might, however, envision such a ‘root tonal morpheme’, e.g., a floating tone sequence $\textcircled{\text{L}}\textcircled{\text{H}}$ meaning ‘eat’ which does not co-occur with any segments, whose tone is realized on neighboring affixes (comparable to segmentally null roots found in certain highly synthetic languages, e.g., Menominee [mez] – Bloomfield 1962: 63).

Tonal morphemes can often be shown to derive from associated tones (but certainly not always), where material on the segmental tier has eroded but tones on the tonal tier remain (the so-called ‘stability’ of tone – Goldsmith 1976, Yip 2002). This may result in allomorphs in synchronic systems. A straightforward example comes from the language isolate Laal [gdm] (Lionnet 2021), which has a series of connective markers which agree with the head of the noun phrase in gender and number (5a). All of these consist of a H-toned CV syllable which may appear as such (first example in 5b–d). In all gender/number combinations except masculine and feminine singular, the connective also has a floating $\textcircled{\text{H}}$ allomorph (second example in 5b–d). The realization of this allomorph on a monosyllabic word results in a rising tone in (5c). As seen in (5d), if the preceding tone is already H then the $\textcircled{\text{H}}$ tonal allomorph does not have a discernible effect on the surface.

- (5) Laal – Connective morphemes show tonal allomorphs (data from Lionnet's field notes)

a. *já* MASC.SG
jí FEM.SG
má NEUT.SG
yá NEUT.PL
yí PL, NHUM.SG

b. *nyàw yí dēē* ~ *nyàw* (H) *dēē* → *nyǎw dēē*
house CON at:me house CON at:me ‘my house’

c. *dōrūm yí dēē* ~ *dōrūm* (H) *dēē* → *dōrūm dēē*
rope CON at:me rope CON at:me ‘my rope’

d. *wúrá yí dēē* ~ *wúrá* (H) *dēē* → *wúrá dēē*
things CON at:me things CON at:me ‘my things’

Markers akin to Laal connectives (also known as ‘associative markers’) are commonly expressed as tonal morphemes, where they act to ‘link’ or ‘associate’ a head noun with either a modifier (often a modifying noun, as in compounds and possession). Examples are pervasive across the African continent (Cahill 2000), including Benue-Congo (e.g., Ebirá [igb] – Adivé 1989: 65–66), Gur (e.g., Dagbani [dag] – Olawsky 1999: 205ff.), Mande (Green 2018), Kru (Marchese 1979: 77), Chadic (Schuh 2017: 141), Nubian (e.g., Midob [mei] – Werner 1993: 24, and Khoe-Kwadi (e.g., Gǀui [gwǀ] – Nakagawa 2006: 60–80.³

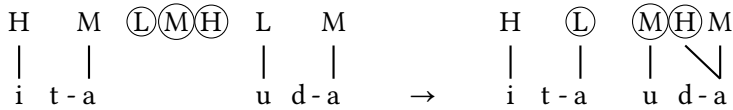
Tonal morphemes may be relatively simple like the $\textcircled{\text{H}}$ tonal allomorph in Laal, but sequences of floating tones exist as well. Consider the language isolate Kunama [kun] (Connell et al. 2000), which has a basic high/mid/low (H/M/L) distinction forming various contours (falling $\widehat{\text{H}}\text{M}$, $\widehat{\text{M}}\text{L}$, $\widehat{\text{H}}\text{L}$, and rising $\widehat{\text{M}}\text{H}$). Compounds and possessive constructions are expressed by juxtaposing two nouns which are linked via a floating tone sequence $\textcircled{\text{L}}\textcircled{\text{M}}\textcircled{\text{H}}$, e.g., in (6) where we gloss it as LINK.

³Outside of Africa, such tonal morphemes are common in noun phrases too, e.g., in Norwegian [nor] (Wetterlin 2010) in Europe, Metnyo Ambel [wgo] (Arnold 2018) and Fore [for] (Scott 1978) in the Pacific, Kuki-Thaadow [tcz] (Hyman 2010: 43) in SE Asia, among others.

- (6) Kunama – Tonal morpheme as floating tone sequence
- | | | | | | |
|----|---------------|---|----------------|---|---------------------------------|
| a. | <i>ít-ā</i> | $\textcircled{L}\textcircled{M}\textcircled{H}$ | <i>úd-ā</i> | → | <i>ít-à úd-a\textcircled{H}</i> |
| | house-SG | LINK | door-SG | | ‘house’s door’ |
| b. | <i>àbìn-ā</i> | $\textcircled{L}\textcircled{M}\textcircled{H}$ | <i>úkùn-ā</i> | → | <i>àbìn-à úkùn-ā</i> |
| | elephant-SG | LINK | ear-SG | | ‘elephant’s ear’ |
| c. | <i>kīĩš-ā</i> | $\textcircled{L}\textcircled{M}\textcircled{H}$ | <i>āsāńg-ā</i> | → | <i>kīĩš-à āsāńg-ā</i> |
| | girl-SG | LINK | head-SG | | ‘girl’s head’ |

In these examples, the $\textcircled{L}$ tone delinks the M of the *-ā* singular suffix of the first (modifying) noun, while the $\textcircled{M}\textcircled{H}$ portion docks to the second noun (the head noun). This creates a $\widehat{HM}$ contour tone on the suffix of in (6a). In contrast, in (6b–c), there is enough segmental material for both $\textcircled{M}$ and $\textcircled{H}$ to be realized on the stem itself, the underlying stem tones being deleted to make room for the floating tones (note that the change of H to M on [ŋ] in /āsāńg-/ → [āsāńg-] in (6c) is caused by an unrelated tonal process). A simple autosegmental representation of (6a) is provided in (7).

- (7) Kunama – Autosegmental diagram of [ít-à úd-a\textcircled{H}] ‘house’s door’



For decades, tonal morphemes have also been the de facto standard in another prosodic area: intonation, especially that of the Autosegmental-Metrical framework (e.g., [Pierrehumbert 1980](#), [Ladd 2008](#)). Under this perspective, one could interpret the rising tone to mark yes/no questions in (Standard) English as a kind of floating tone sequence, since it enters the phonological derivation unassociated to any specific element on the segmental tier. Even many tone languages have such intonational tonal morphemes to mark specific types of clauses, e.g., the ‘African lax question prosody’ whereby utterance-final L tones commonly indicate yes/no questions especially in the tonally-rich middle belt of Africa ([Rialland 2009](#)).

2.2 Floating tone as part of morphemes

There is also evidence that floating tones can form part of the underlying representation of morphemes, where they accompany segmental material. Consider another Mande language, Guinean Kpelle [gkp]. In this language, nouns pronounced with all L tones form two classes, exemplified by *yàlà* ‘lion’ and *gbònò*

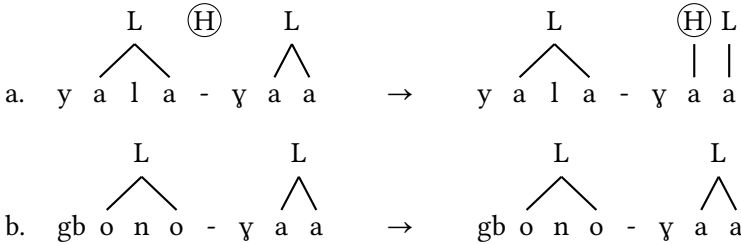
‘ring’ in (8). The latter has no unexpected tonal impacts on its environment, while the former systematically adds a H tone to the following morpheme, both within and across words (Konoshenko 2008: 24).

(8) Guinean Kpelle – Distinct tonal behaviors of L-toned nouns

- a. *yàlà* ‘lion’ *yàlà-yàà* ‘lions’ *ɛ̃ yàlà káà* ‘I saw a lion’
 b. *gbònò* ‘ring’ *gbònò-yàà* ‘rings’ *ɛ̃ gbònò kàà* ‘I saw a ring’

A straightforward way to capture the above difference is to posit a floating tone in the underlying representation of /*yàlà*($\textcircled{\text{H}}$)/ ‘lion’. This ($\textcircled{\text{H}}$) associates to a following L tone syllable if one exists but is otherwise deleted. Autosegmental diagrams of the plural forms with *-yàà* are in (9).

(9) Guinean Kpelle – Autosegmental diagrams of the plural forms with *-yàà*



Konoshenko (2008: 26–27) shows that the *yàlà*-type is realized with M rather than L tone in closely related varieties – e.g., *yālā* ‘lion’ in Liberian Kpelle [xpe] contrasting against *gbònò* ‘ring’ – and posits that this difference derives historically from a *c̣ṿc̣ṿ vs. *c̣ṿc̣ṿ contrast (cf. Hyman 2003). This represents a running theme in floating tones: they diachronically derive from previously associated tones which become dislodged from their original position.

Moreover, in many languages the same sequence of tones can contrast depending on whether they are floating or associated in the underlying representation. A three-way contrast is found in the Grassfields Bantu language Bamileke-Ghomala’ [bbj] (Nissim 1981) where nouns may have a LH sequence pre-associated to the segmental tier (10a), or can be a L followed by a floating ($\textcircled{\text{H}}$) (10b) or a H preceded by a floating ($\textcircled{\text{L}}$) (10c). These three patterns contrast against a simple linked H tone (10d). (Note that the $\downarrow$ indicates downstep, and the degree sign $^\circ$ indicates a non-falling L tone before pause.)

(10) Bamileke-Ghomala' – Contrast between associated and floating tones

(Nissim 1981: 153, 156, 249)

- | | | | | | | | |
|----|------|---------------|---|-------------|----------|------------------------------------|----------------------|
| a. | LH | <i>bvǎ</i> | → | <i>bvǎ</i> | 'dog' | <i>thá</i> [↓] <i>bvǎ</i> | 'head of the dog' |
| b. | L(H) | <i>bàp(H)</i> | → | <i>bàp°</i> | 'animal' | <i>thá</i> <i>bàp°</i> | 'head of the animal' |
| c. | (L)H | <i>(L)dhá</i> | → | <i>dhá</i> | 'spouse' | <i>thá</i> [↓] <i>dhá</i> | 'head of the spouse' |
| d. | H | <i>fá</i> | → | <i>fá</i> | 'parent' | <i>thá</i> <i>fá</i> | 'head of the parent' |

For maximum clarity, we provide autosegmental diagrams for these four contrasts, in (11).

(11) Bamileke-Ghomala' – Autosegmental diagrams of floating tone contrasts

- | | | | | | | | |
|----|------|----|-------|----|-------|----|-----|
| | L H | | L (H) | | (L) H | | H |
| | ∨ | | | | | | |
| a. | bv ǎ | b. | b a p | c. | dh ə | d. | f á |

The exact evidence for these floating tones involves both their surface pronunciation and the tonal effects they have on their environment. We turn to this in §3.

Just as floating tone can be part of the representation of lexical morphemes, it can also co-accompany grammatical morphemes, e.g., affixes and clitics. In Chichewa [nya] (Hyman & Mtenje 1999), there are two otherwise homophonous derivational suffixes which appear in the same post-verbal field: toneless causative *-its* versus intensive *-its(H)* with a floating tone, in (12).⁴

(12) Chichewa – Homophonous suffixes with and without (H) Hyman & Mtenje 1999: 95, 98

- | | | | |
|----|------------------------------|---|---------------------------------------|
| a. | No derivational suffixation: | | |
| | <i>thandiz-a</i> | → | <i>thandiz-a</i> 'help!' |
| b. | Causative suffixation: | | |
| | <i>thandiz-its-a</i> | → | <i>thandiz-its-a</i> 'cause to help!' |
| c. | Intensive suffixation: | | |
| | <i>thandiz-its(H)-a</i> | → | <i>thandiz-its-á</i> 'help a lot!' |

Such grammatical floating tones are very common in the derivational and inflectional morphology of African (and other) languages.

In languages with floating tones, typically they only appear on a minority of morphemes, whereas most morphemes do not have floating tones. However, in

⁴These forms are further subject to penultimate lengthening, H tone anticipation, and default L insertion (cf. Hyman & Mtenje 1999: 128, endnote 7 and references therein; for a recent description and analysis of the phonology of Chichewa, see Downing & Mtenje 2017).

the Ijoid language Izon [ijc], Rolle (2021) posits that virtually all lexical items sponsor floating tones, which dock to the following morpheme (and frequently override that morpheme's tone). Representative examples of the 'ubiquity' of floating tone are in (13), which show four different patterns on nouns: either a $\textcircled{L}$ or $\textcircled{H}$ floating tone alone (13a–b), or a $\textcircled{L}\textcircled{H}$ or $\textcircled{H}\textcircled{L}$ sequence (13c–d) (association of these floating tones is sometimes vacuous, as in 13c).

- (13) Izon – Representative examples of ubiquitous floating tone on virtually all lexical items

a.	Ends in $\textcircled{L}$	<i>wún</i> $\textcircled{L}$	<i>ẹ́rì</i>	→	[wún ẹ̀rì]	'see sand!'
b.	Ends in $\textcircled{H}$	<i>fù</i> $\textcircled{H}$	<i>ẹ́rì</i>	→	[fù ẹ́rì]	'see salt!'
c.	Ends in $\textcircled{L}\textcircled{H}$	<i>tà</i> $\textcircled{L}\textcircled{H}$	<i>ẹ́rì</i>	→	[tà ẹ́rì]	'see wife!'
d.	Ends in $\textcircled{H}\textcircled{L}$	<i>wò</i> $\textcircled{H}\textcircled{L}$	<i>ẹ́rì</i>	→	[wò ẹ́rì]	'see him!'

Another potential case of ubiquitous floating tone concerns languages with stem or word 'tone melodies' which predictably map onto the corresponding tone-bearing units (see Hyman 2003 for a brief history of the concept of 'tone melody' in African linguistics). Mende [men] is perhaps the most famous language in this regard (Leben 1973, Leben 1978), but a clearer case of predictable tone association (with fewer counterexamples) comes from the Bantu language Kukuya [kkw] (Paulian 1975, Hyman 1987). This language has exactly five tone melodies L, H, LH, HL, and LHL. These map onto the stem constituent in a predictable fashion, shown in (14) (association ignores prefixes, which are outside of the stem).

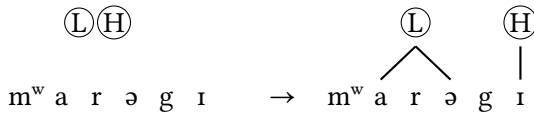
- (14) Kukuya – Five tone melodies (on stem)

a.	L	<i>(kì-)</i> bàà	<i>(kì-)</i> bàlàgà
		'jealousy'	'to change route'
b.	H	<i>(mà-)</i> báá	<i>(lì-)</i> bálágá
		'cheeks'	'fence'
c.	LH	<i>(mò-)</i> sàá	<i>m^wàràgí</i>
		'seed necklace'	'younger brother'
d.	HL	<i>(kì-)</i> káà	<i>(kì-)</i> kàràgà
		'to grill'	'to be entangled'
e.	LHL	<i>(ndé-)</i> kàây	<i>(ndé-)</i> kàlógì
		'he loses weight'	'he turns around'

In such a language, words with the LH melody such as *m^wàràgí* 'younger brother' have only one association possibility (trisyllabic L.L.H) and thus cannot

contrast against words of the shapes L.H.H, L.L.LH, etc. Tonal melodies are not traditionally analyzed as floating tones, because their association is predictable and systematic, and the trivial step of association is unnecessary to include in the underlying representation. However, strictly speaking, if tonal melodies are not associated in underlying representations, this is equivalent to being floating tones, shown in (15) (cf. also McPherson 2022).

- (15) Kukuya – Predictable association of tone melody as ‘floating tones’



3 Floating tone effects

As has been established, the primary reason for proposing floating tones is to account for tonal patterns which are conditioned by the presence of a morpheme and are not predicted based simply on the tones that are realized on that morpheme or any other morpheme around it. Such patterns we collectively refer to as ‘floating tone effects’, the empirical substance motivating a floating tone analysis.

The most straightforward of these is the presence of a tonal alternation on a neighboring morpheme, e.g., the fact that Guinean Kpelle *yàlà* ‘lion’ causes a following L to become H as in *yàlà-yàà* ‘lions’ (cf. *gbònò-yàà* ‘rings’ in example 8 above). In this section, we shall show additional floating tone effects in three interrelated patterns: blocking effects, downstep and upstep/raising (i.e., register effects), and final level Ls.

3.1 Blocking effects

A widespread property of tone is its ability to spread onto neighboring environments, e.g., a H spreading one position to the right. When two tonally identical morphemes should, according to the rules of the language, spread their H tone, but only one of them does, one common analysis is to posit that there is a covert floating tone which ‘blocks’ spreading.

Such effects are common in the Grassfields languages of Cameroon. For example, in Aghem [agq] (Hyman 1979, 2003), H-toned nouns can be split into different groups based on their tonal effects, though they are pronounced identically

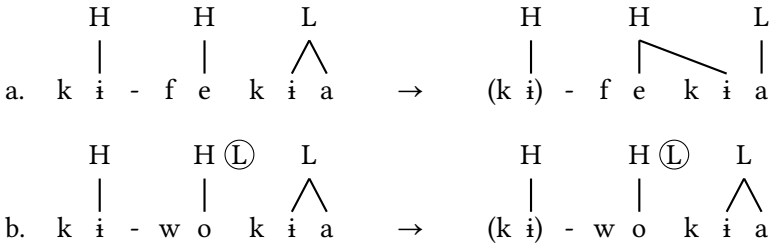
in isolation. In one type of H words, the H spreads onto a following L tone word (16a), whereas in another type of H words there is no tone spreading (16b).⁵

(16) Aghem – Two types of H-toned nouns

- a. *kí-fé* ‘leg’ (*kí-*)*fé kíà* ‘your (sg) leg’
 b. *kí-wó* ‘hand’ (*kí-*)*wó kíà* ‘your (sg) hand’

This contrast falls out naturally in an analysis with floating tones: the first type ends in a H which spreads one position to the right (17a), but the second type has a final floating $\textcircled{\text{L}}$ which blocks this spreading operation (17b). (Additional evidence for the presence of this floating $\textcircled{\text{L}}$ is given in the next section.)

(17) Aghem – Blocking effect from floating tone



3.2 Downstep and upstep/raising

Another common motivation for floating tones is to explain otherwise unpredicted register changes which lower or raise the pitch range, i.e., ‘downstep’ and ‘upstep’.⁶ We can demonstrate downstep using the Aghem data from Section 3.1: /*kí-fé*/ ‘leg’ vs. /*kí-wó* $\textcircled{\text{L}}$ / ‘hand’ with a floating tone. When these nouns appear before a H-toned modifier such as *kín* ‘this’ (agreeing in noun class), they again show different tonal patterns. Without a floating tone, both the noun and the modifier are realized at the same H tone level (18a), whereas when there is a floating $\textcircled{\text{L}}$ the H tone of the modifier is realized within a lower register, indicated by the downstep mark ↓ (18b).

⁵In this example the two nouns have an underlying /H/ prefix which is deleted before most modifiers. See the original sources for details.

⁶Note that downstep and upstep do not always result from the effect of a floating tone, but may also be caused by associated tones, as in the automatic downstepping (‘downdrift’) of a H tone by a preceding associated L tone, or cases where a H is automatically downstepped after another H (cf. Stewart 1965, 1983; Connell 2011; Snider, this volume, among others). Downstep may even be a phonological primitive present in underlying representation, independent of tone, e.g., in the African language Baga Pukur [bcg] (Rochant 2023; cf. also Lionnet 2022, 2025, on the Oceanic languages Paici [pri], Drubea [duf], and Numèè [kdk] of New Caledonia).

(18) Aghem – Two types of H-toned nouns

- a. /kí-fé kín/ → (kí-)fé kín ‘this leg’
 b. /kí-wó(̤) kín/ → (kí-)wó ↓kín ‘this hand’

As emphasized by Hyman (2003: 160), by positing a single floating tone we can simultaneously account for both the downstep pattern in (18b) and the blocking effect in (16b). That there are two pieces of evidence of a floating ̤ at the end of the root /wó(̤)/ shows that the analysis is not circular or ad hoc. We will return to this argument in Section 5, where we discuss alternatives.

An interesting case of floating tones in sequence is found in Bamileke-Medumba [byv], another Grassfields language (Voorhoeve 1971). In Medumba, nouns which are pronounced as H in isolation can correspond to different underlying forms: a simple underlying /H/ (19a), a H with a preceding floating tone /̤H/ (19b), and a H with a preceding and following floating tone /̤H̤/ (19c). They are differentiated when they combine in a ‘noun of noun’ genitive construction linked only by a floating (̤) tonal morpheme (another example of this linking function, cf. Section 2.1). When there is only a H tone on both sides of the tonal morpheme (̤) (19d), the result is a level H tone across the phrase (the initial floating tone of ̤tí ‘tree’ has no effect here). In contrast, the final floating tone of ̤yú(̤) causes downstep of any H tone which follows, in (19e). What is remarkable is the case of (19f), where the tonal morpheme (̤) is flanked by ̤ tones: the second noun ‘child’ is realized on a double downstepped level, a greater interval than a single downstep.

(19) Medumba – Effects of multiple floating tones

- a. /sán/ → sán ‘bird’
 b. /̤tí/ → tí ‘tree’
 c. /̤yú(̤)/ → yú ‘thing’
 /̤mén(̤)/ → mén ‘child’
 d. /̤tí + (̤) + sán/ → tí sán ‘tree of bird’
 e. /̤yú(̤) + (̤) + sán/ → yú ↓sán ‘thing of bird’
 f. /̤yú(̤) + (̤) + ̤mén(̤)/ → yú ↓↓mén ‘thing of child’

One can understand this as the ̤ before the second noun causing the first downstep and the ̤ before the tonal morpheme (̤) as causing the second downstep, resulting in a H ↓(̤) ↓H sequence. The downsteps then ‘combine’ to form the [H ↓↓H] surface form.⁷

⁷Another option for (19f) is to elongate the vowel of ‘thing’ and produce a falling contour with successive downsteps, i.e., yú↓ú ↓mén (Voorhoeve 1971: 48).

Although floating L tones often trigger downstep, they occasionally instead condition a preceding ‘upstep’ whereby the register is raised rather than lowered. One example comes from Ebira [igb] (Scholz 1976), a language with a three-way H/M/L distinction. In this language, H tones (including sequences of H tones) are automatically raised to super-high (S) when they precede a L tone, both within words (20a) and across words (20b).⁸

(20) Ebira – Change of H to super-high (S) before a L-toned syllable (Scholz 1976: 67)

- | | | | | |
|----|--------------|---|--------------|-----------------------------|
| a. | /àgùgù/ | → | [àgùgù] | ‘crocodile’ |
| b. | /ínómí nìní/ | → | [ínómí nìní] | ‘birds’ (i.e., bird PLURAL) |

This upstep/raising process applies opaquely in cases where there is a latent L tone which does not appear on the surface. Positing a floating ① in these contexts unifies the triggering environments: the change to S applies before any kind of L tone on the tonal tier (whether associated or not). The floating ① may be present in the underlying representation (where it minimally contrasts in the same position with words without floating tone – 21a and 21b), or it may be a derived floating ① which comes from a pre-associated L tone which becomes delinked due to operations like H tone spreading (21c) and vowel elision (21d).

(21) Ebira – Change of H to S before floating ① (Scholz 1976: 41,63,68,71,72,82)

- | | | | | |
|----|------------|---------|-----------|-----------------------|
| a. | /hámá wá/ | → | [hámá wá] | ‘imitate them!’ |
| | /cémá① wá/ | → | [cěmǎ wá] | ‘lift them!’ |
| b. | /ōzí émī/ | → | [ōzí émī] | ‘my son’ |
| | /ōzí ①yọ/ | → | [ōzí yọ] | ‘how is the child?’ |
| c. | /ó-wèyí/ → | ó-wé①yí | → | [ǒwéyí] ‘a small one’ |
| d. | /dọ òré/ → | d-ó①ré | → | [dǒré] ‘get glass’ |

3.3 Final level L tone

A final type of floating tone effect comes from languages which contrast two different types of pre-pausal L tones: one L which descends to the bottom of one’s pitch range (transcribed simply L) versus another L which has a level pitch and does not descend as such (transcribed as L° with a degree symbol). Under a

⁸Given that the tones following the raised tone go unaffected by this raising process (as in 20b), it may be more appropriate to simply call this ‘H-raising’ rather than ‘upstep’. Our main point remains, however: this same process is triggered equally by associated and floating L tones.

floating tone analysis, the level L° is due to a final floating tone ($\textcircled{H}$ or $\textcircled{M}$) which blocks the utterance-final fall of the ordinary L.

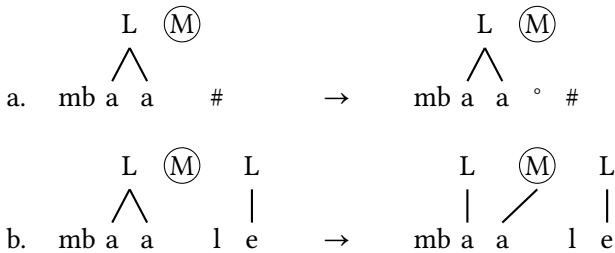
Yet another Grassfields language can demonstrate this pattern, this time from Bamileke-Fe'fe' [fmp] (Hyman 1972) with a basic L/M/H contrast. Example (22a) demonstrates a contrast between two types of Ls. The noun *pàà* 'wing' in isolation is pronounced with descending pitch before pause, a cross-linguistically widespread process of final downgliding. In contrast, the noun *mbàà* 'meat' has the low level pitch without this downgliding. In context (22b), we see that 'wing' stays L when followed by the proximal demonstrative *lè* 'this', while 'meat' is realized with a $\widehat{LM}$ rising contour.

(22) Fe'fe' – Two types of floating Ls (Hyman 1972: 131–139)

- a. *pàà* 'wing' *mbàà* 'meat'
 b. *pàà* *lè* 'this wing' *mbàà* *lè* 'this meat'

We can account for this contrast by positing a final floating $\textcircled{M}$ on 'meat'. When followed by a L tone, this $\textcircled{M}$ docks to the preceding word resulting in a contour, but utterance-finally it is not realized, and simply blocks the application of final L 'downgliding' (i.e., only L's which are absolutely final on the tonal tier undergo this process). Although $\textcircled{M}$ does not associate per se, it still affects the surface pronunciation, as shown in the autosegmental representations in (23).

(23) Fe'fe' – Floating $\textcircled{M}$ blocks final L 'downgliding'



4 Typological parameters

Once floating tones are assumed, they become a phonological object which we can evaluate like any other less-abstract object. In this section, we explore several typological issues which arise with floating tones, having to do with inventory (what tones can float and in which contexts), locality (how locally does the floating tone associate), prosodic status (can tonal morphemes pattern as affixes, cli-

tics, or even independent words), directionality (do floating tones dock leftwards or rightwards), and markedness (what phonological conditions favor/disfavor tone docking).

4.1 Inventory

Thus far, we have seen many different kinds of floating tones. This includes various qualities such as floating $\textcircled{\text{H}}$'s, $\textcircled{\text{L}}$'s, and even $\textcircled{\text{M}}$'s, appearing in various combinations to form a complex floating tone sequence such as $\textcircled{\text{H}}\textcircled{\text{L}}$ in Izon (13), $\textcircled{\text{L}}\textcircled{\text{H}}\textcircled{\text{M}}$ in Kunama (6), etc. At the same time, we saw that floating tones may appear on their own as 'tonal morphemes' or can come before or after a segmental sponsor (sometimes both – Medumba in 19c). We can thus ask: are there any limits on a language's inventory of floating tones?

One possibility is that only tones which can be associated can be floating. In other words, if we find floating $\textcircled{\text{H}}$ underlyingly, we will also find underlying associated H. While tempting, there are in fact analyses which posit that certain tone qualities must come in floating. We call these 'inherently floating tones'. This is how Daly & M. (2007) characterize the contrast between H and L in the Mexican language Peñoles Mixtec [mil], where these two tones systematically behave differently. To account for this asymmetry, Daly and Hyman propose to analyze L tones as inherently floating – i.e., all H tones are underlyingly linked to a segment, while all L tones enter the derivation floating. While this remains an analytical possibility, this has not been proposed to account for any African tonal language to date.

The closest cases of inherently floating tone we can think of in Africa come from grammatical tone patterns in three languages. One is Fe'Fe', already seen in Section 3.3 above, which has a lexical M vs. L distinction, but an additional H tone contrast is found only in grammatical contexts. For example, *vhū* 'ashes', *mōh* 'fire', and *mūū* 'child' are underlyingly M, but are replaced by a tonal morpheme $\textcircled{\text{H}}$ in connective constructions, e.g., *vhú mōh* 'ashes of fire', *vhú mōh mūū* 'ashes of fire of child' (Hyman 1976: 127–128). If this tonal morpheme is interpreted as floating, then this would entail that all H tonemes come in floating. Two additional examples are Kisi [kss] (Childs 1995) and Igala [igl] (Ejeba 2016), which both have a super-high (S) tone that occurs only in certain grammatical contexts, primarily negation. In Kisi, negation is expressed as a change to S on the verb plus a final particle *lé* NEG (24), while in Igala negated clauses the tone on subject clitics (SCL) becomes S co-accompanied by the clitic *=ń* NEG (25).⁹

⁹A reviewer brings up an alternative that Igala S is derived from a floating low (comparable

- (24) Kisi – Super-high tone (S) restricted to grammatical tone patterns (Childs 1995: 49)

a.	à	dàtá	yá	b.	à	dàtǎ	yá	lé
	you	condemn	me		you	condemn.NEG	me	NEG
	‘You condemned me.’				‘You didn’t condemn me.’			

- (25) Igala – S toneme restricted to negation (Ejeba 2016: 326)

òdʒòrú=ǔ	dʒē	èg ^w à=ń
Ojoru=NEG.SCL	eat	beans=NEG
‘Ojoru did not eat beans.’		

Another potential type of restriction is not whether certain tones can float, but whether they are permitted to float if they are adjacent to an associated identical tone. For example, one could imagine a language with a ban on H($\bar{H}$) (a ban in line with the ‘Obligatory Contour Principle’). While most of the examples we have encountered incidentally comply with such a restriction, we cannot hold this to be a true universal. Indeed, we have seen comparable sequences, e.g., in the Izon word *tà*($\bar{L}$)($\bar{H}$) ‘wife’ (13c), whereby a linked L is followed by a floating $\bar{L}$ (see Rolle 2021 for the original arguments). Another case comes from Dagbani (Hyman 1993: 242), where monosyllabic H nouns are interpreted as having a floating ($\bar{H}$) suffix in order to account for an unexpected limitation in a spreading rule, e.g., *bá*-($\bar{H}$) ‘father’.

In total, while it is certainly the case that an individual language may impose a restriction on its inventory of floating tones or the types of environments they may appear adjacent to, with rare exceptions, floating tones are usually a subset of associated tones.

4.2 Locality

In the examples we have seen thus far, floating tone has a local effect: it docks to an adjacent syllable/morpheme, it triggers downstep or raising of an adjacent tone, or it blocks an adjacent tone from triggering a tonal operation (e.g., spreading, or final L ‘downgliding’). In general, floating tones typically target local environments. This is only a strong tendency, however, as non-local floating tone patterns are attested as well (depending on one’s interpretation), usually under specific circumstances.

to Ebira in 21). Regardless, the Igala data demonstrates our main point about tonal inventories, irrespective of the chosen analysis.

Non-local linking of floating tones can be the result of ‘prominence-based association’. In this type, a floating tone associates to a phonologically-prominent target in the relevant domain, such as to stress or to a particular boundary (or a fixed position relevant to that boundary). In order to qualify as prominence-based association, we must find evidence independent from floating tone association for this position being prominent. One example comes from the Bantu language Zanzibar Simakonde [kde] (Manus 2014), hereafter simply Makonde. As is common in Eastern Bantu, the penultimate position of the phrase can be identified as prominent since it undergoes penultimate vowel lengthening. Manus demonstrates that pre-verbal tense/aspect/mood (TAM) prefixes can co-vary with one of four floating tone patterns: $\underline{\text{L}}$, $\underline{\text{H}}$, $\underline{\text{L}}\underline{\text{H}}$, or $\underline{\text{L}}\underline{\text{H}}\underline{\text{L}}$. While the exact pattern is idiosyncratic to the TAM prefix, where they appear is predictable: they are always realized on the penultimate syllable, the position of prominence. In the examples in (26), the penultimate position shows both penultimate lengthening and floating tone association (data from Manus 2014: 267–268). The origin of the floating tones and its surface position are underlined.¹⁰

(26) Makonde – Prominence-based association of floating tones

- | | | | |
|----|-------------------|--|----------------------|
| a. | CONSECUTIVE | /ni- <u>ka</u> $\underline{\text{H}}$ - | takatukil-a/ |
| | ‘and I stood up’ | ni- <u>ka</u> - | takatuk <u>i</u> ĩla |
| b. | FUTURE | /ni- <u>nda</u> $\underline{\text{L}}\underline{\text{H}}$ - | takatukil-a/ |
| | ‘I will stand up’ | ni- <u>nda</u> - | takatuk <u>i</u> ĩla |
| c. | REMOTE PAST | /ní- <u>ndi</u> $\underline{\text{L}}\underline{\text{H}}\underline{\text{L}}$ - | takatukil-a/ |
| | ‘I stood up’ | ní- <u>ndi</u> - | takatuk <u>i</u> ĩla |
| | 1SG- TAM- | | stand.up-INFL |

Of course, as with all uses of the floating tone concept, it is debatable whether floating tones are the best way to capture these data (as opposed to the tones being pre-linked to a specific affix and then delink and move to the penult). However, our main point remains: the segmental component of the TAM category appears non-local to the tonal component.

Not all non-local floating tone association is due to prominence, however. Consider another Bantu language, Luganda [lug] (Hyman & Katamba 2010), which contrasts H vs. L vs. Ø (toneless) on syllables/morphemes. Consider the example in (27) (in the underlying forms in this example, we denote H with an acute

¹⁰Note that the H tone on the 1SG prefix *ní-* in (26c) which spreads onto the following syllable constitutes a separate type of grammatical process, independent of those floating tones attracted to the penultimate position. See Manus (2014: 268ff.) for details.

accent on its host vowel, L with a grave accent, and Ø as unmarked).¹¹

(27) Luganda - Non-local tone association across clitics (Hyman & Katamba 2010: 95)

- a. /te-y-a-túùk-à/ (output of lexical phonology)
 tè-yà-túùk-à (surface output)
 NEG-3SG-arrive-FV
 ‘s/he didn’t arrive’
- b. /te-y-a-túùk-à (H)bùli lunákù/
 tè-yà-túùk-à bùli lùnákù
 NEG-3SG-arrive-FV every day
 ‘s/he didn’t arrive every day’
- c. /te-y-a-túùk-à na=byaa=(H)bùli mu-limi/
 tè-yà-túùk-à nà=byàà=bùli mù-límí
 NEG-3SG-arrive-FV with=those.of=every CL-farmer
 ‘s/he didn’t arrive with those of every farmer’
- d. /te-y-a-túùk-à na=byaa=mu-limi/
 tè-yà-túùk-à nà=byàà=mù-límí
 NEG-3SG-arrive-FV with=those.of=CL-farmer
 ‘s/he didn’t arrive with those of the farmer’

Some words have an initial floating (H), such as (H)bùli ‘every’ which appears initial within its noun phrase. While the verb occurs with final L in (27a), the floating (H) of ‘every’ is realized on the final syllable of the verb in (27b). (We underline the relevant positions in the example.) Example (27c) demonstrates that even when the trigger (H)bùli is separated by one or more proclitics (here *na*= ‘with’ and *byaa*= ‘those of’), it still assigns its tone non-locally to the final position of the verb. Example (27d) shows that these clitics do not otherwise result in a H tone on the verb.

An even more extreme example of non-prominence-based non-local association comes from the counting rules of Kuria [kuj] (Mwita 2008, Marlo et al. 2015). As in Makonde in (26) above, inflectional prefixes idiosyncratically sponsor one of several floating tone patterns. However, unlike in Makonde these do not associate to the same position; rather, each of them targets a different position within the verbal stem. Depending on the grammatical context, a floating tone may be assigned to the first ((H)¹ in 28b), second ((H)² in 28c), third ((H)³ in 28d), of fourth

¹¹We refer the reader to the original source for details on how toneless syllables get tone on the surface (e.g., involving spreading, boundary tones, among other factors).

mora of the stem ($\textcircled{\text{H}}^4$ in 28e), and even to the first and fourth simultaneously ($\textcircled{\text{H}}^1\textcircled{\text{H}}^4$ in 28f). An example set is in (28) using the verb *kirigiit*- ‘scrub’, which forms a stem constituent labeled ‘ Σ ’. As above, the position that floating tones associate to is underlined.¹²

(28) Kuria – Contrastive association of floating tones (Mwita 2008: 305–336)

- | | | | |
|----|---|---|--|
| a. | $\emptyset$ – HORTATORY IMPERATIVE
/to-tá-(kirigiit-a) Σ / | → | <i>to-tá-(kirigiit-a)Σ</i>
‘let us scrub’ |
| b. | $\textcircled{\text{H}}^1$ – HABITUAL PAST
/n-to-ogá $\textcircled{\text{H}}^1$ -(kirigiit-ire) Σ / | → | <i>n-to-ogá-(kírigiit-ire)Σ</i>
‘we used to scrub (then)’ |
| c. | $\textcircled{\text{H}}^2$ – PAST PROGRESSIVE (FOCUS)
/n-to-oga $\textcircled{\text{H}}^2$ -(kirigiit-ire) Σ / | → | <i>n-to-oga-(kírígíit-ire)Σ</i>
‘(indeed) we have scrubbed (today)’ |
| d. | $\textcircled{\text{H}}^3$ – REMOSTE FUTURE (FOCUS)
/n-to-re $\textcircled{\text{H}}^3$ -(kirigiit-a) Σ / | → | <i>n-to-re-(kirigíit-a)Σ</i>
‘we will scrub (then)’ |
| e. | $\textcircled{\text{H}}^4$ – INCEPTIVE
/to-ra $\textcircled{\text{H}}^4$ -(kirigiit-a) Σ / | → | <i>to-ra-(kirigíit-a)Σ</i>
‘we are about to scrub’ |
| f. | $\textcircled{\text{H}}^1\textcircled{\text{H}}^4$ – NARRATIVE PAST
/to-gá $\textcircled{\text{H}}^1\textcircled{\text{H}}^4$ -(kirigiit-a) Σ / | → | <i>to-gá-(kírígíit-a)Σ</i>
‘(and) we scrubbed’ |

As extensively argued in Marlo et al. (2015), there is no evidence that any of these positions is particularly prominent in Kuria (and some are rare or otherwise unattested cross-linguistically, i.e., $\textcircled{\text{H}}^3$ and $\textcircled{\text{H}}^4$).

In the above non-local patterns, there is no tone intervening between the origin of the floating tone and its final surface position. Cases of an associated tone being transparent to non-local association are extremely rare. One case outside of Africa is the Mexican language San Miguel el Grande Mixtec [mig] (Mak 1950, Zimmermann 1950) whereby a floating $\textcircled{\text{H}}$ which normally docks locally skips over the first tone under exceptional conditions (e.g., $\textcircled{\text{H}}$ + ML → M $\textcircled{\text{H}}$). However, in most examples in Africa (as worldwide), non-locality requires that there be no intervening tones.¹³

¹²Note that the actual surface forms in Kuria differ slightly from what is provided in (28), due to orthogonal tone spreading rules, not shown.

¹³One possible instance of non-local association over an intervening tone is in Tommo So [dto]

4.3 Prosodic status

Segmental grammatical morphemes are often catalogued as to how tightly they ‘cohere’ with their environment, what we call their ‘prosodic status’. Morphemes may be tightly bound to their base as affixes, loosely bound as clitics, or not bound at all as independent function words. If ‘tonal morphemes’ as defined in Section 2.1 are just like regular segmental morphemes, only without any segmental content, then we should expect them to demonstrate the same variety of prosodic statuses. Such issues of prosodic status are unfortunately rarely discussed with respect to tonal morphemes and floating tones in general.

One overt attempt at assessing different kinds of tonal morphemes explicitly with these prosodic categories in mind is Van de Velde’s (2008, 2009) analysis of Eton [eto]. Van de Velde starts from a simple logical extension: since tonal rules are sensitive to different kinds of boundaries (e.g., affix, clitic, and word), the tonal behavior of floating tones should reveal the prosodic status (affix, clitic, or word) of the tonal morphemes they are part of. That is, it should not matter whether the tones that undergo tonal rules are underlyingly associated with segments or not, and we should be able to find affixes, clitics, and independent grammatical words that consist only of tonal material. What Van de Velde shows is that floating tones do indeed have three distinct sets of patterns, which correspond exactly to the patterns found with tones unequivocally associated with segmental affixes, clitics, and independent grammatical words.

To illustrate the first type, let us look at the augment prefix (used on nouns in certain contexts). This prefix has several allomorphs based on the noun class of the noun to which they attach, some of which consist of a vowel and a H tone, as in class 7 /i-/ in (29a’, b’), while others consist only of a floating (H) tone, as is the case for nouns of class 5 illustrated in (29a, b). As can be seen in (29a, b), the floating (H) of the augment docks to a following L-toned syllable and thereafter spreads rightward onto all following L-toned syllables to form a HL contour tone with the last L in the sequence. This directly mirrors the behavior of the H tone from the segmentally-overt allomorphs of the augment prefix in (29a’, b’).

(McPherson 2014, McPherson & Heath 2016), though the original sources expressly do not adopt a floating tone analysis for the relevant tonal patterns. See these sources and Rolle (2018: 30–31, 81–82) for details.

- (29) Eton – Tonal morpheme as a prefix / $\textcircled{\text{H}}$ -/
- | | | | | |
|-----|-------------------------------------|---------|---|--------------------------------|
| a. | / $\textcircled{\text{H}}$ -è-dʒóŋ/ | dì | → | <u>é</u> -dʒóŋ dì |
| | AUG5-CL5-hole | CL5.DEM | | ‘this hole’ |
| a’. | /í-bì-sínà / | bí | → | <u>í</u> - <u>bí</u> -sínà bí |
| | AUG7-CL7-hole | CL7.DEM | | ‘these cats’ |
| b. | / $\textcircled{\text{H}}$ -è-dʒôŋ/ | dì | → | <u>é</u> -dʒôŋ dì |
| | AUG5-CL5-clan | CL5.DEM | | ‘this clan’ |
| b’. | /í-bì-lòlì / | bí | → | <u>í</u> - <u>bí</u> -lwàlì bí |
| | AUG7-CL5-clan | CL7.DEM | | ‘these ducks’ |

In contrast, consider two other tonal morphemes: those that act as a connector CON= in a ‘noun of noun’ construction (30), and those that act as a locative preposition (30). These are analyzed as a proclitic and as a word, respectively. Both have several allomorphs, consisting of either a (C)V syllable with an associated H tone (examples a’ and b’ in 30 and 31), or only a floating $\textcircled{\text{H}}$ tone (examples a and b). The floating $\textcircled{\text{H}}$ allomorph of each of these two morphemes shows distinct behavior compared to the prefixal $\textcircled{\text{H}}$ seen in (29a–b). Firstly, while in both cases, the $\textcircled{\text{H}}$ is realized on the following position (on the class prefix in all examples in 30 and 31), it does not spread any further after association. Compare the falling tone in the surface form of the root in (29b) above to the L tone on the root in (30b) and (31b) below. Independent word tonal morphemes further condition downstep when they are followed by another H tone (31a), which is not seen with either prefixal $\textcircled{\text{H}}$ (29a) or proclitic $\textcircled{\text{H}}$ (30a). In each of these cases, the behavior of the tonal morpheme mirrors the tonal behavior of its segmental allomorph.

- (30) Eton – Tonal morpheme as a proclitic / $\textcircled{\text{H}}$ =/
- | | | | | |
|-----|-------------|------------------------------------|---|-----------------------------|
| a. | /bj-ǎ | $\textcircled{\text{H}}$ =bì-sínà/ | → | bj-ǎ <u>bí</u> -sínà |
| | CL8-claw | CL8.CON=CL8-cat | | ‘the claws of the cats’ |
| a’. | /bj-ǎ | bí=ì-sínà / | → | bj-ǎ b= <u>í</u> -sínà |
| | CL8-claw | CL8.CON=CL8-cat | | ‘the claws of a cat’ |
| b. | /mà-vùl | $\textcircled{\text{H}}$ =bì-lòlì/ | → | mà-vùl <u>bí</u> -lwàlì |
| | CL6-feather | CL6.CON=CL7-duck | | ‘the feathers of the ducks’ |
| b’. | /mà-vùl | má=ì-lòlì/ | → | mà-vùl m= <u>í</u> -lwàlì |
| | CL6-feather | CL6.CON=CL7-duck | | ‘the feathers of a duck’ |

- (31) Eton – Tonal morpheme as an independent word (H)
- a. / $\textcircled{\text{H}}$ è-dʒóŋ/ → $\underline{\acute{\epsilon}}\text{-}\downarrow dʒóŋ$
 LOC CL5-hole ‘in the hole’
- a’. /á mǝ-dʒóŋ/ → á $\underline{mǝ}\text{-}\downarrow dʒóŋ$
 LOC CL6-hole ‘in the holes’
- b. / $\textcircled{\text{H}}$ è-dʒòŋ/ → $\underline{\acute{\epsilon}}\text{-}dʒòŋ$
 LOC CL5-clan ‘in the clan’
- b’. /á mǝ-dʒòŋ/ → á $\underline{mǝ}\text{-}dʒòŋ$
 LOC CL6-clan ‘in the clans’

We summarize the three sets of tonal patterns in Table 1.

Table 1: Three types of tonal morphemes in Eton

		L.L	L.H	Spreading after association	Downstep
Prefix	/ $\textcircled{\text{H}}$ -/	[$\textcircled{\text{H}}$. $\textcircled{\text{H}}$ L]	[$\textcircled{\text{H}}$.H]	Yes	No
Proclitic	/ $\textcircled{\text{H}}$ =/	[$\textcircled{\text{H}}$.L]	[$\textcircled{\text{H}}$.H]	No	No
Word	/ $\textcircled{\text{H}}$ /	[$\textcircled{\text{H}}$.L]	[$\textcircled{\text{H}}$. $\downarrow$ H]	No	Yes

Given the parallels with their segmental counterparts, extending the affix/clitic/word distinction to tonal morphemes appears to be warranted for Eton. However, there exist other cases where floating tonal morphemes show distinct tonal behavior (e.g., triggering vs. not triggering spreading), but where a comparable prosodic categorization as in Eton is without support (we will turn to one of these shortly in Section 4.5 from Giphende). We thus emphasize that just as analyzing tonal morphemes as floating tones is an analysis, so is their prosodic status and the constituents they form.

4.4 Directionality

Another parameter of floating tones is directionality: A floating tone may dock to the right or to the left of its postulated origin depending on a number of factors. The easiest way to see directionality is to show how it applies differently in different varieties of a language. Consider the case of Igbo where a floating $\textcircled{\text{H}}$ tone functions to link two nouns in a ‘noun of noun’ construction (Hyman

& Schuh 1974; see also Section 2.1 above). In Central Igbo [ibo], the $\textcircled{\text{H}}$ tonal morpheme docks leftward to the end of the first noun (32a), while in Aboh Igbo [ukw] it docks rightward to the beginning of the second noun (32b). This is consistent with Aboh having regular H tone spreading (*ézi* → [ézi] ‘pig’), which Central Igbo does not (cf. [ézi]).

(32) Igbo – Directionality of association of tonal morpheme

- a. Central Igbo: *àgbà* + $\textcircled{\text{H}}$ + *èṅwè* → *àgbà* *èṅwè* ‘jaw of monkey’
- b. Aboh Igbo: *ègbà* + $\textcircled{\text{H}}$ + *èṅwè* → *ègbà* *èṅwè* ‘jaw of monkey’

The exact conditions on directionality vary greatly from language to language, and even family to family. When generalizing over floating tone directionality across Eastern Grassfields Bantu, Hyman & Tadadjeu (1976: 62) find several general criteria, summarized in (33).

(33) Eastern Grassfields Bantu – Criteria for floating tone directionality:

- a. Realization: Associate in the direction that has greater tonal effect on surface
e.g., for an input L $\textcircled{\text{H}}$ H, associating to the left would have a greater effect than associating to the right where the output would be surface-identical to L H
- b. Boundary strength: Do not associate in a direction that crosses certain boundaries
e.g., for an input L $\textcircled{\text{H}}$ [L (where ‘[’ indicates a stem boundary), associating to the left would not cross the stem boundary while associating to the right would
- c. Naturalness: Associate in the direction of a more phonologically natural outcome
e.g., for an input $\widehat{\text{HL}}$ $\textcircled{\text{H}}$ L involving a falling $\widehat{\text{HL}}$ contour, associating to the left would create a marked $\widehat{\text{HLH}}$ falling-rising contour which is worse than associating to the right creating two falling contours

4.5 Markedness

In short, a number of factors are potentially involved, and the way floating tones dock can only be partially predicted. This leads into our last parameter: phonological markedness.

One of the great insights of phonology is that sequences of segments, tones, and their combinations can be assessed as to whether they are relatively ‘marked’

or ‘unmarked’ in terms of complexity or frequency, e.g., a word *spear* in English is less marked than *sphere* with an unusual [sf] onset. Particularly marked structures can be actively avoided in languages and structures with floating tones are no exception.

Consider the case of Ik [ikx], a Kuliak language of Uganda with a H vs. L tonal contrast (Schrock 2014: 121, 598). This language shows depressor consonant effects, whereby the pitch lowering effects of voiced obstruents has been phonologized. Consider the association of a floating (H) which appears with the subordinating conjunction clitic *na=* CONJ ‘if/when’. The floating tone associates to the syllable immediately following *na=* if that syllable begins with a sonorant or voiceless obstruent (i.e., non-depressors, 34a–b), but falls on *na=* itself if the following syllable begins with a voiced obstruent (i.e., a depressor, 34c) (L tones are not transcribed in the examples).

(34) Ik – Effect of depressor consonants on floating tone association

- a. /na(H)=ɲk-ese dád-a dakú-gwarí-oó/ → *na=ɲk-ese* ...
 CONJ=eat-SPS honey-NOM tree-top-ABL
 ‘if honey is eaten from the tree top’
- b. /na(H)=kək-ese ... → *na=kók-ese* ...
 CONJ=close-SPS ‘if (SUBJECT) is tied...’
- c. /na(H)=gon-ese ... → *ná=gon-ese* ...
 CONJ=look-SPS ‘if (SUBJECT) is looked at...’

In Ik, a floating tone docks rightward except if it would create a marked syllable (i.e., a H tone with a depressor consonant), in which case it docks leftward.¹⁴

It is just as common, however, to find floating tones insensitive to depressor consonant effects. And just as we saw in §4.3, floating tones even in the same language may not be equally sensitive to markedness conditions. One illustrative example comes from the Bantu language Giphende [pem] (Hyman 2017: 113–114). In Giphende, nouns have an underlying tone which is systematically changed depending on its syntactic position, e.g., /gì-kómbó/ ‘broom’ with underlying /L-H-H/ (where /gì-/ is a noun class prefix) is pronounced [gì-^lkómbó] when used as a syntactic object, [gì-kómbó] as a focused object or in a genitive, and [gì-kómbó] in predicative position (‘it’s a ’). Hyman divides these into four ‘tonal

¹⁴That the floating (H) tone can dock to the morpheme *na=* which introduces it is an instance of ‘self-association’. In many languages, self-association is itself marked and actively avoided (a markedness constraint which goes by various names in the literature – see McPherson 2014: 89, and Rolle 2018: 258 for discussion). Another kind of markedness is thus whether or not a floating tone can dock to the morpheme which sponsors it.

cases' in (36). We label these as FOCUS.OBJECT (FOC), GENITIVE (GEN), ACCUSATIVE (ACC), and PREDICATIVE (PRED) for our purposes, realized as the tonal morphemes labeled $\textcircled{H}_F$, $\textcircled{H}_G$, $\textcircled{H}_A$, and $\textcircled{H}_P$ respectively.¹⁵

These four morphemes can be assessed as to which conditions allow them to dock, and when they dock what their behavior is afterwards. As per the first aspect, FOCUS.OBJECT $\textcircled{H}_F$ only docks to all L-toned nouns, GENITIVE $\textcircled{H}_G$ docks as long as it would not be adjacent to a H, while ACCUSATIVE $\textcircled{H}_A$ and PREDICATIVE $\textcircled{H}_P$ always dock. And as per the second aspect, $\textcircled{H}_F$ systematically spreads in its only context (with all-L nouns), $\textcircled{H}_G$ and $\textcircled{H}_P$ only spread in this all-L context but otherwise do not spread. Finally, accusative $\textcircled{H}_A$ always spreads over L tones up to the next H, which it causes to downstep. This is summarized in Table 2.

Table 2: Tonal case in Giphende

Underlying tone of noun	Case $\textcircled{H}_F$ FOC	$\textcircled{H}_G$ GEN	$\textcircled{H}_A$ ACC	$\textcircled{H}_P$ PRED
L-L.L	$\textcircled{H}\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}.\textcircled{L}$
L-L.L.L	$\textcircled{H}\textcircled{H}\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}\textcircled{H}.\textcircled{L}$
L-L.H	L-L.H	$\textcircled{H}\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}.\downarrow\textcircled{H}$	$\textcircled{H}\textcircled{H}.\textcircled{L}$
L-L.H.L	L-L.H.L	$\textcircled{H}\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}.\downarrow\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}.\textcircled{L}$
L-H.L	L-H.L	L-H.L	$\textcircled{H}\textcircled{H}.\downarrow\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}.\textcircled{L}$
L-H.H	L-H.H	L-H.H	$\textcircled{H}\textcircled{H}.\downarrow\textcircled{H}.\textcircled{H}$	$\textcircled{H}\textcircled{H}.\textcircled{H}$
L-H.H.L	L-H.H.L	L-H.H.L	$\textcircled{H}\textcircled{H}.\downarrow\textcircled{H}.\textcircled{H}.\textcircled{L}$	$\textcircled{H}\textcircled{H}.\textcircled{H}.\textcircled{L}$
L-H.H.H	L-H.H.H	L-H.H.H	$\textcircled{H}\textcircled{H}.\downarrow\textcircled{H}.\textcircled{H}.\textcircled{H}$	$\textcircled{H}\textcircled{H}.\textcircled{H}.\textcircled{H}$

We can understand these Giphende patterns as different strategies for avoiding two different H tones on the tonal tier within the same domain (in formal terms, avoiding a violation of the ‘Obligatory Contour Principle’). $\textcircled{H}_F$ FOC is the strictest and only associates if there is no other H in the word, $\textcircled{H}_G$ GEN is less strict in that it can dock as long it would not be adjacent to another H, whereas $\textcircled{H}_A$ ACC and $\textcircled{H}_P$ PRED are the least strict in allowing the floating tone to dock in any context (with idiosyncratic spreading and downstep effects thereafter to solve any violation of the ‘Obligatory Contour Principle’). Sensitivity to the above and other markedness considerations is thus not a single parameter which can be set for an entire language, but rather is calibrated for each individual floating tone or tonal morpheme.

¹⁵See the original source (Hyman 2017) for details on the exact distribution of these tonal morphemes, which depend on factors such as negation, dislocation, relative clausehood, etc.

5 Alternatives to floating tones

We have presented a plethora of examples where floating tones have proven to be useful in accounting for otherwise unexplained tonal patterns, and in the last section we showed how they can be parameterized along several phonological dimensions, just like less abstract phonological objects such as segments or associated tones. In this section, we shall explore four families of alternatives to floating tones: 1) positing a distinct string of pre-associated tones, 2) endowing a tone with a diacritic, 3) adjusting the ‘strength’ or ‘gradience’ of a tone, or 4) assuming universal allomorphy.

To demonstrate each of these alternatives, let us return to the Ghomala’ data from Section 2.2 (Nissim 1981: 150, 153). In Table 3, we provide a six-way tonal contrast using only H and L tonemes. Under a floating tone approach, there is a critical contrast between LH vs. L(H) vs. (L)H, which contrast against associated H, L, and HL.

Table 3: Floating tones and their alternatives (numbered 1 through 4) in Ghomala’

	[fá] ‘parent’	[tà] ‘pot’	[bûə] ‘madman’	[bvǔ] ‘dog’	[bàp°] ‘animal’	[dhó] ‘spouse’
Floating tone analysis	H /fá/	L /tà/	HL /bûə/	LH /bvǔ/	L(H) /bàp(H)/	(L)H /Ldhó/
1) Different Tones	H fá	L tà	HL bûə	LH bvǔ	M bàp	MH dhə̃
2) Diacritic	H1 fá	L1 tà	HL bûə	LH bvǔ	L2 bàp	H2 dhó
3) Strength	H _[1.0] fá	L _[0.5] tà	HL bûə	LH bvǔ	L _[1.0] bàp	H _[0.5] dhó
4) Allomorphy	/fá/	/tà/ /ᵀtà/	/bûə/	/bvǔ/ /ᵀbvú/	/bàp°/ /ᵀbàp°/	/dhó/ /ᵀdhó/

For Ghomala’, floating tones were posited to capture effects not observed with their counterparts without floating tones, e.g., the contrast between /fá/ ‘parent’ and /Ldhó/ ‘spouse’ in (35) within ‘noun of noun’ constructions, where the floating (L) of the latter triggers downstep in (35b).

(35) Ghomala' – Floating tones posited to capture downstep in 'noun of noun' construction

- a. thó fá → thó fá 'the head of the parent'
 b. thó $\textcircled{L}$ dhó → thó $\downarrow$ dhó 'the head of the spouse'

Under the first alternative analysis in Table 3, words with erstwhile floating tones actually constitute different tones or tone sequences, i.e., the form for 'animal' could be /bāp/ with a M tone rather than /bāp $\textcircled{H}$ / with a floating tone, while 'spouse' could be /dhə $\textcircled{L}$ / with a MH contour tone rather than / $\textcircled{L}$ dhó/. This would avoid floating tones, but necessitate both more tone heights (a H vs. L. vs. M contrast) as well as particular realizational rules which map /MH/ to surface [H] in isolation but as [$\downarrow$ H] after a H tone.

Under the second alternative, we would posit two qualitatively different types of H and L tones endowed with a diacritic, e.g., H vs. H2 and L1 vs. L2. Tonal rules would then be sensitive to this distinction, such that the trigger or target of a rule could target both or only one. For example, H1 would not downstep after another H tone, but H2 would downstep (as in 35b above). As argued in Hyman (2003), what this diacritic solution misses is the fact that floating tone effects act 'as if' a tone was there, i.e., we posit an abstract / $\textcircled{L}$ / or / $\textcircled{H}$ / because it patterns like a concrete /L/ or /H/. There is nothing explicit like this in the diacritic approach.¹⁶

In the third, from the Gradient Symbolic Representation literature (Smolensky & Goldrick 2016, Zimmermann 2018, 2019, Hsu 2022), floating tone effects would involve qualitatively identical tonemes without any diacritics, but they would bear different 'weights' or 'strengths'. For example, the H tone on 'parent' would be a 'strong' H tone with a weight of [1.0] (i.e., /fá_[1.0]/) while the H on 'spouse' would be a 'weak' one with a weight of perhaps [0.5] (i.e., /dhó_[0.5]/). The former tones would not undergo any downstep when they follow another H (i.e., they are strong enough to resist any tone change), while the latter tones would show downstep (i.e., they are not strong enough to resist tone change). The main challenge of this approach is explicitly connecting the 'strength' of a tone to the behavior it shows—and, again, accounting for the parallel behavior of $\textcircled{L}$ s and Ls.

Finally, the last and alternative to floating tones posits pure allomorphy. One recent proposal is the Emergent Phonology model of Archangeli & Pulleyblank (2022) which eschews underlying representations for 'morph sets', essentially turning all phonological alternations into a matter of allomorph selection. To

¹⁶For further argumentation against diacritic approaches, see, *inter alia*, Clements & Ford (1977, 1979), summarized in Lionnet & Hyman (2018).

demonstrate this model, consider the Yoruba [yor] data in (36). Such data are conventionally understood as the result of a productive and language-general rule whereby the H in L-H sequences becomes rising due to L tone spread, i.e., the verb in the example has one underlying form /rí/ which surfaces as [rí] or [rĩ] depending on whether tonal spreading takes place. In contrast, under Emergent Phonology this verb would consist of the morph set /rí/, /rĩ/ where neither is the underlying representation of the other. The allomorph /rí/ is selected after a H-toned word whereas the allomorph /rĩ/ is selected after a L-toned word.

- (36) Yoruba – Tonal alternations as stored allomorphs in a morph set {rí, rĩ}
 a. ó rí ‘he/she/it saw’ b. kò rĩ ‘he/she/it did not see’

Emergent Phonology is a self-described ‘surface-to-surface model’, which has no conceptual place (or need) for floating tones (39). The contrastive floating tone effects in (41) would be captured by distinct morph sets, e.g., the ‘parent’ type we analyzed as /H/ would constitute a morph set only consisting of H-toned {fá}, while the ‘spouse’ type we analyzed as /ᐁH/ would constitute a larger morph set {dhá, ᐁdhá}, and so on.

In comparing these five possible analyses, one can see that floating tones, contrary to the other four alternatives, are able to capture multiple tonal phenomena simultaneously. This gives floating tones a clear advantage over the alternatives, and explains why they have been part and parcel of tonological work (especially in Africa). This was seen in Aghem (Section 3) where we posited a contrast between /kí-fé/ ‘leg’ with no floating tone versus /kí-wóᐁ/ ‘hand’. The floating ᐁ conditions downstep on a following H tone as well as blocking H tone spread onto a following L tone. Crucially, we know independently that an overt (associated and realized) L tone has the exact same effects. Our contention is that any adequate analysis should capture both of these patterns using the same machinery.

Moreover, one of the most intuitive motivations for floating tones concerns the very existence of tonal morphemes. Whether one adopts different tones, quantitative diacritics, or qualitative strength, tonal morphemes (and hence floating tones) exist in underlying representations. Of the alternatives, only the Emergent Phonology model could capture floating tone effects without literal tonal morphemes, by assuming one morph in a neutral context (e.g., /tà/) and another in a specific grammatical context (e.g., /tǎ/), which would correspond to a floating tone effect. We will not further evaluate the merits and shortcomings of Emergent Phonology in this paper, but merely point out that if it is adopted, the phonological module would look so radically different that even entertaining the idea

of floating tones becomes illogical. Although we disprefer these alternatives for capturing floating tone effects, they may prove successful in individual cases. That is, one may wish to posit distinct pre-associated tone patterns rather than floating tones for an individual tonal effect, and stored allomorphy rather than a tonal operation involving association of floating tones to account for some other tonal effect.

6 Summary

In this chapter, we have overviewed the use and function of floating tones, demonstrating their integral part in both the description and formal analysis of African languages. We showed that floating tones can be part of morphemes or act as independent morphemes themselves. Their function as an analytic device is to account for certain otherwise unpredicted tonal behavior, what we called ‘floating tone effects’. Examples include tone patterns on surrounding morphemes, as well as blocking effects, downstep and upstep/raising, and contrastive final level low tones. Once floating tones are treated as *bona fide* phonological objects, a number of analytic issues arise. Some of those include its inventory in a language, as well as locality, prosodic status, directionality, and markedness. Throughout this chapter we emphasized that floating tones are an analysis with several alternative analyses available, although none was found to have the range and flexibility of the floating tone construct.

In conclusion, abstract floating tones and their function as purely tonal morphemes are parallel to their less abstract segmental counterparts in virtually all details. They can occur as an affix or as a clitic, depending on whether it comes in at the word or phrase level, and may either be added outside the tone of its base or host, or it may replace it entirely. A floating tone may be the sole exponent of a morphological expression, or it may co-occur with another morph in a type of bipartite construction, e.g. a tense designation defined as a tonal prefix plus a segmental suffix. Ultimately, the most important take-away is that if one wishes to eschew floating tones entirely, any alternative must capture the full range of floating tone effects which we have outlined in this chapter.

Abbreviations

AUG	augment (followed by class number when relevant)
CL	noun class (followed by class number when relevant)
CON	connective
CONJ	conjunction
FV	final vowel
NHUM	non-human
PRED	predicative
SCL	subject clitic (in Igala)
SPS	sequential impersonal passive (in Ik)

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References

- Adivé, John R. 1989. *The verbal piece in Ebira*, vol. 85 (Summer Institute of Linguistics: Publications in Linguistics). Dallas: Dallas, Texas: The Summer Institute of Linguistics & the University of Texas at Arlington.
- Archangeli, Diana & Douglas Pulleyblank. 2022. *Emergent phonology*. Berlin: Language Science Press.
- Arnold, Laura. 2018. Lexical tone in Metnyo Ambel. *Oceanic Linguistics* 57(1). 199–220.
- Bloomfield, Leonard. 1962. *The Menomini language*. New Haven, Conn.: Yale University Press.
- Cahill, Michael. 2000. Tonal associative morphemes in Optimality Theory. In *Osu working papers in linguistics*, vol. 53, 31–70. Ohio State University.
- Childs, G. Tucker. 1995. *A grammar of Kisi: a Southern Atlantic language*. Berlin: Mouton de Gruyter.
- Clements, George N. & Kevin C. Ford. 1977. On the phonological status of downstep in kikuyu. *Harvard Studies in Phonology* 1. 187–272.
- Clements, George N. & Kevin C. Ford. 1979. Kikuyu tone shift and its synchronic consequences. *Linguistic Inquiry* 10. 179–210.

- Connell, Bruce. 2011. Downstep. In Marc van Oostendorp, Colin Ewen, Elizabeth Hume & Keren Rice (eds.), *The Blackwell companion to phonology*, 824–847. Oxford: Blackwell Publishing.
- Connell, Bruce A., Richard J. Hayward & John Abrahama Ashkaba. 2000. Observations of Kunama tone (Barka dialect). *Studies in African linguistics* 29(1). 1–41.
- Daly, John P. & Hyman, Larry M. 2007. On the representation of tones in Peñoles Mixtec. *International Journal of American Linguistics* 73. 165–208.
- Downing, Laura J. & Al Mtenje. 2017. *The phonology of chichewa* (Oxford linguistics). Oxford University Press.
- Eberhard, David M., Gary F. Simons & Charles D. Fennig (eds.). 2023. *Ethnologue: languages of the world. twenty-second edition*. Dallas, Texas: SIL International. <http://www.ethnologue.com>.
- Ejeba, Salem Ochala. 2016. *A grammar of Igala*. University of Port Harcourt. (Doctoral dissertation).
- Goldsmith, John. 1976. *Autosegmental phonology*. Massachusetts Institute of Technology. (Doctoral dissertation).
- Green, Christopher. 2018. A survey of word-level replacive tonal patterns in Western Mande. *Mandenkan* 59. 1–35.
- Hsu, Brian. 2022. Gradient symbolic representations in Harmonic Grammar. *Language and Linguistics Compass* 16(9). e12473.
- Hyman, Larry. 1993. Structure preservation and postlexical tonology in Dagbani. In Sharon Hargus & Ellen M. Kaisse (eds.), *Studies in Lexical Phonology*, 235–254. San Diego, CA: Academic Press.
- Hyman, Larry M. 1972. *A phonological study of Fe'fe' Bamileke*. University of California, Los Angeles. (Doctoral dissertation).
- Hyman, Larry M. 1976. D'où vient le ton haut du bamileke-fe'fe'? In Larry M. Hyman, Leon C. Jacobson & Russell G. Schuh (eds.), *Papers in african linguistics in honour of william e. welmers*, vol. Suppl. 6 to Studies in African linguistics (Suppl. 6 to Studies in African linguistic), 123–134. Los Angeles: African Studies Center & Department of Linguistics, University of California Los Angeles.
- Hyman, Larry M. 1979. *Aghem grammatical structure*. Larry M. Hyman (ed.), vol. 7 (Southern California Occasional Papers in Linguistics). Los Angeles: Department of Linguistics, University of Southern California.
- Hyman, Larry M. 1987. Prosodic domains in Kukuya. *Natural Language and Linguistic Theory* 5. 311–333.
- Hyman, Larry M. 2001. Tone systems. In Martin Haspelmath, Ekkehard König, Wulf Oesterreicher & Wolfgang Raible (eds.), *Language typology and language universals: an international handbook*, vol. 2, 1367–1380. New York & Berlin: Walter de Gruyter.

- Hyman, Larry M. 2003. African languages and phonological theory. *Glott International* 7(6). 153–163.
- Hyman, Larry M. 2010. Kuki-Thaadow: an African tone system in Southeast Asia. In Franck Floricic (ed.), *Essais de typologie et de linguistique générale: mélange offerts à Denis Creissels*, 31–51. Lyon: Presses de l'Ecole normale supérieure.
- Hyman, Larry M. 2017. Disentangling conjoint, disjoint, metatony, tone cases, augments, prosody, and focus in Bantu. In Jenneke van der Wal & Larry M. Hyman (eds.), *The conjoint/disjoint alternation in Bantu*, 100–121. Berlin: Mouton De Gruyter.
- Hyman, Larry M. & Francis X Katamba. 2010. Tone, syntax, and prosodic domains in Luganda. *UC Berkeley PhonLab Annual Report* 6(6). 21–42.
- Hyman, Larry M. & Al Mtenje. 1999. Prosodic morphology and tone: the case of Chichewa. In René Kager, Harry van der Hulst & Wim Zonneveld (eds.), *The prosody-morphology interface*, 90–133. Cambridge: Cambridge University Press.
- Hyman, Larry M. & Russell Schuh. 1974. Universals of tone rules: evidence from West Africa. *Linguistic Inquiry* 5. 81–115.
- Hyman, Larry M. & Maurice Tadadjeu. 1976. Floating tones in Mbam Nkam. In Larry M. Hyman (ed.), *Studies in Bantu tonology* (Southern California Occasional Papers in Linguistics 3), 57–111. Los Angeles: University of Southern California.
- Konoshenko, Maria. 2008. Tonal systems in three dialects of the Kpelle language. *Mandenkan* 44. 21–42.
- Ladd, D. Robert. 2008. *Intonational phonology*. Cambridge: Cambridge University Press.
- Leben, Will. 1978. The representation of tone. In Victoria Fromkin (ed.), *Tone: a linguistic survey*, 177–219. New York: Academic Press.
- Leben, William R. 1973. *Suprasegmental phonology*. Cambridge, MA: Massachusetts Institute of Technology. (Doctoral dissertation).
- Lionnet, Florian. 2021. The gender system of Laal. *STUF - Language Typology and Universals* 74(2). 241–262. DOI: <https://doi.org/10.1515/stuf-2021-1031>.
- Lionnet, Florian. 2022. Tone and downstep in Paici (Oceanic, New Caledonia). *Phonological Data and Analysis* 4(1). 1–47. DOI: <https://doi.org/10.3765/pda.v4art1.45>.
- Lionnet, Florian. 2025. Tonal languages without tone: downstep in Drubea and Numèè (Oceanic, New Caledonia). *Phonology* 42. e23. DOI: <https://doi.org/10.1017/S0952675725100171>.

- Lionnet, Florian & Larry M. Hyman. 2018. Current issues in african phonology. In Tom Güldemann (ed.), *The languages and linguistics of Africa*, vol. 11 (The World of Linguistics (Hans Henrich Hock ed.)), 602–708. Berlin: Mouton de Gruyter. DOI: <https://doi.org/10.1515/9783110421668-006>.
- Maddieson, Ian. 2022. Le ton, est-il originel?–le lien musique-langage. In *Proc. xxxive journées d'études sur la parole–jep 2022*, 933–942.
- Maddieson, Ian. 2023. Tone is not predominant: tone is not primordial. In *Proceedings of the 20th international congress of phonetic sciences (icphs 2023)*, 1925–1929.
- Mak, Cornelia. 1950. A unique tone perturbation in Mixteco. *International journal of American linguistics* 16(2). 82–86.
- Manus, Sophie. 2014. Melodic patterns in Símákonde. *Africana Linguistica* 20(4). 263–276.
- Marchese, Lynell. 1979. *Atlas linguistique kru: essai de typologie*, vol. 73 (Linguistique africaine). Abidjan: Université d'Abidjan: Institut de Linguistique Appliquée.
- Marlo, Michael R., Leonard Chacha Mwita & Mary Paster. 2015. Problems in Kuria h tone assignment. *Natural Language and Linguistic Theory* 33(1). 251–265.
- McPherson, Laura. 2014. *Replacive grammatical tone in the Dogon languages*. University of California, Los Angeles. (Doctoral dissertation).
- McPherson, Laura. 2022. Word tone is epiphenomenal. In Haruo Kubozono, Junko Ito & Armin Mester (eds.), *Prosody and prosodic interfaces*, 3–28. Oxford: Oxford University Press.
- McPherson, Laura & Jeffrey Heath. 2016. Phrasal grammatical tone in the Dogon languages. *Natural Language and Linguistic Theory* 34(2). 593–639.
- Mwita, Leonard Chacha. 2008. *Verbal tone in Kuria*. Los Angeles: University of California at Los Angeles. (Doctoral dissertation).
- Nakagawa, Hiroshi. 2006. *Aspects of the phonetic and phonological structure of the G/ui language*. Johannesburg: University of Witwatersrand. (Doctoral dissertation).
- Nissim, Gabriel M. 1981. *Le bamileke-ghomálá' (parler de Banjoun, Cameroun): phonologie, morphologie nominale, comparaison avec des parlers voisins* (Tradition orale 45). Paris: SELAF.
- Olawsky, Knut J. 1999. *Aspects of Dagbani grammar with special emphasis on phonology and morphology*, vol. 41 (LINCOM Studies in African Linguistics). München: Lincom.
- Paulian, Christiane. 1975. *Le kukuya: langue teke du Congo*. Paris: SELAF.

- Pierrehumbert, Janet. 1980. *The phonetics and phonology of English intonation*. Cambridge, MA: Massachusetts Institute of Technology. (Doctoral dissertation).
- Pulleyblank, Douglas. 1986. *Tone in lexical phonology*. Dordrecht: D. Reidel.
- Rialland, Annie. 2009. The African lax question prosody: its realisation and geographical distribution. *Lingua* 119(6). 928–949.
- Rochant, Neige. 2023. *Bilectal grammar of Baga Pukur, an Atlantic language of Guinea*. Paris: Université Sorbonne Nouvelle. (Doctoral dissertation).
- Rolle, Nicholas. 2018. *Grammatical tone: typology and theory*. University of California, Berkeley. (Doctoral dissertation).
- Rolle, Nicholas. 2021. Lexical tone contrast in Izon as ubiquitous floating tone. *Phonological Data and Analysis* 3(4). 1–40. DOI: [10.3765/pda.v3art4.37](https://doi.org/10.3765/pda.v3art4.37).
- Scholz, Hans-Jürgen. 1976. *Igbira phonology*. Microfiche ed., vol. 7 (Language Data Africa Series). Dallas: SIL. 114.
- Schrock, Terrill. 2014. *A grammar of Ik (Icé-tód): northeast Uganda's last thriving Kuliak language*. Utrecht: Rijksuniversiteit te Leiden. (Doctoral dissertation).
- Schuh, Russell G. 2017. *A Chadic cornucopia*. Los Angeles: University of California Los Angeles Department of Linguistics. <https://escholarship.org/uc/item/5zx6z32d> (30 November, 2025).
- Scott, Graham. 1978. *The Fore language of Papua New Guinea*, vol. 47 (Pacific Linguistics: Series B 47). Canberra: Research School of Pacific & Asian Studies, Australian National University.
- Smolensky, Paul & Matthew Goldrick. 2016. Gradient symbolic representations in grammar: the case of French liaison. Ms. available as ROA 1286 in the Rutgers Optimality Archive.
- Stewart, John M. 1965. *The typology of the Twi tone system*. Preprint from the Bulletin of the Institute of African Studies 1, Institute of African Studies, University of Ghana, Legon, Ghana.
- Stewart, John M. 1983. Downstep and floating low tones in Adioukrou. *Journal of African Languages and Linguistics* 5(1). 57–78.
- Van de Velde, Mark. 2008. *A grammar of Eton*, vol. 46 (Mouton Grammar Library). Berlin: Mouton de Gruyter.
- Van de Velde, Mark L. O. 2009. Eton tonology and morphosyntax: a holistic typological approach. In Patience Epps & Alexandre Arkhipov (eds.), *New challenges in typology: transcending the borders and refining the distinctions*, 35–60. Berlin: De Gruyter.
- Voorhoeve, Jan. 1971. Tonology of the Bamileke noun. *Journal of African Languages* 10(2). 44–53.

- Welmers, William. 1959. Tonemics, morphotonemics, and tonal morphemes. *General Linguistics* 4. 1–9.
- Welmers, William. 1973. *African language structures*. Berkeley: University of California Press.
- Welmers, William E. 1976. *A grammar of Vai*, vol. 84 (University of California Publications in Linguistics). Berkeley: Berkeley & Los Angeles: University of California Press.
- Werner, Roland. 1993. *Tìdn-áal: a study of Midob (Darfur-Nubian)*, vol. 17 (Sprache und Oralität in Afrika: Frankfurter Studien zur Afrikanistik). Berlin: Dietrich Reimer.
- Wetterlin, Allison. 2010. *Tonal accents in norwegian*. Berlin: de Gruyter.
- Yip, Moira. 2002. *Tone*. Cambridge: Cambridge University Press.
- Zimmermann, Eva. 1950. The power of a single representation: morphological tone and allomorphy. *Morphology* 26(3–4). 269–294.
- Zimmermann, Eva. 2018. Being exceptional is being weak: tonal exceptions in San Miguel el Grande Mixtec. In Gillian Gallagher, Maria Gouskova & Sora Heng Yin (eds.), *Proceedings of the 2017 Annual Meeting on Phonology*. Linguistic Society of America. DOI: <http://dx.doi.org/10.3765/amp>.
- Zimmermann, Eva. 2019. Gradient symbolic representations and the typology of ghost segments. In Katherine Hout, Anna Mai, Adam McCollum, Sharon Rose & Matthew Zaslansky (eds.), *Proceedings of the 2018 Annual Meeting on Phonology*, 1–12.